Appendix C - Abrolhos to Eastern Recherche Data Analysis Report

South-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A. Candidate predictors were depth, roughness and detrended bathymetry (Geoscience Australia 250 m bathymetry), and long-term means of sea surface temperature (SST), sea level anomaly (SLA), eastward (U) and northward (V) geostrophic current velocity and net primary productivity (PP; MODIS VGPM using OC3 chlorophyll), all resampled to the 250 m bathymetry grid. Survey years were pooled, and zone status (No-Take or Fished) was included in every model.

Benthic ecosystem extent

Table C1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Benthic samples are pooled across survey years, so survey year was not offered as a candidate predictor. Zone status was included in every candidate model.

Response	Model	Δ AICc	ω AICc	R^2	EDF
Macroalgae	**detrended+Z+SLA+V+status**	**0**	**1**	**0.60605**	**10.00**
Sand	**detrended+roughness+SST+PP(OC3)+status**	**0**	**1**	**0.43373**	**10.00**
Seagrass	**detrended+SST+SLA+PP(OC3)+status**	**0**	**1**	**0.6522**	**9.25**
Rock	**Z+SST+SLA+U+status**	**0**	**1**	**0.34578**	**9.97**
Sessile invertebrates	**roughness+Z+SST+V+status**	**0**	**1**	**0.29307**	**9.96**
Reef	**detrended+roughness+SST+U+status**	**0**	**1**	**0.45223**	**8.75**

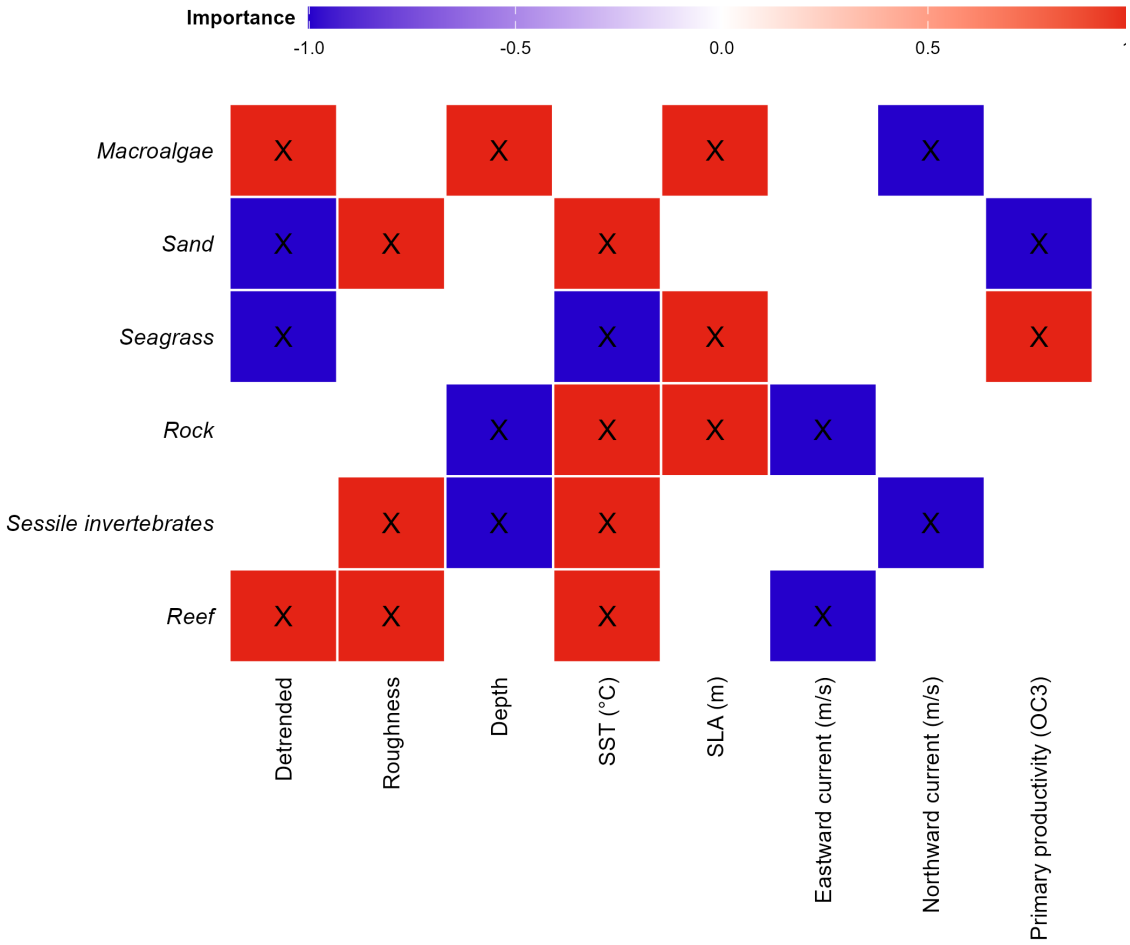


Figure C1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships.

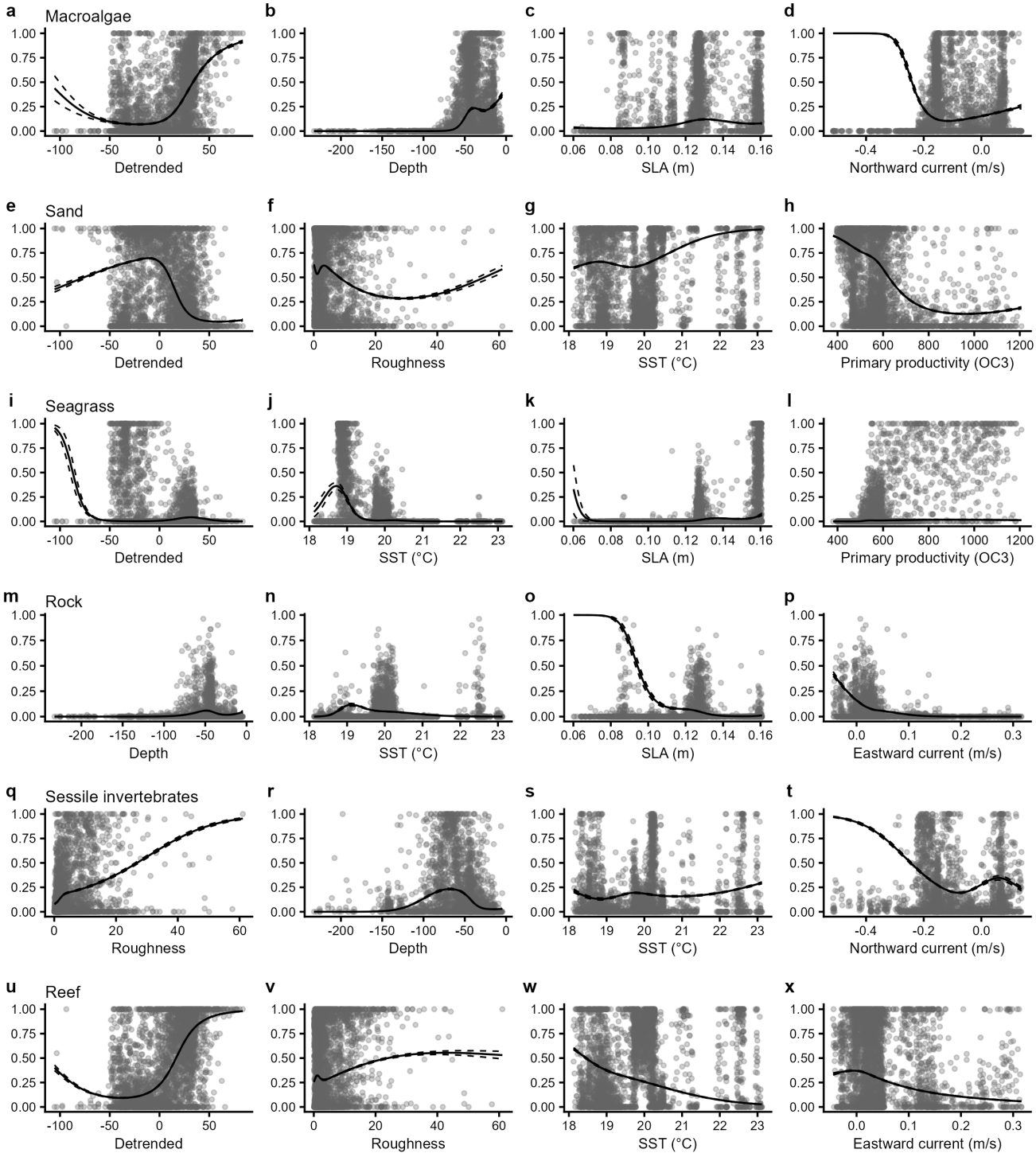


Figure C1.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Benthic samples from 2014, 2020, 2021, 2022, 2023, 2024 and 2025 are pooled, because the campaigns sample different areas and a survey-year term would read as a spatial difference rather than change through time. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data.

Demersal fish assemblage

Table C1.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays. Survey years are pooled and zone status was included in every candidate model.

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
Species richness	**roughness+Z+U+reef+status**	**0**	**0.937**	**10603.36**	**0.43032**	**8.70**
Total abundance	**Z+U+reef+status**	**0**	**0.735**	**19024.10**	**0.23405**	**7.50**
Large Reef Fish Index*	**SST+PP(OC3)+reef+status**	**0**	**1**	**33025.98**	**0.12753**	**7.31**
Reef Fish Thermal Index	**SST+PP(OC3)+reef+status**	**0**	**0.955**	**4518.73**	**0.31194**	**7.73**

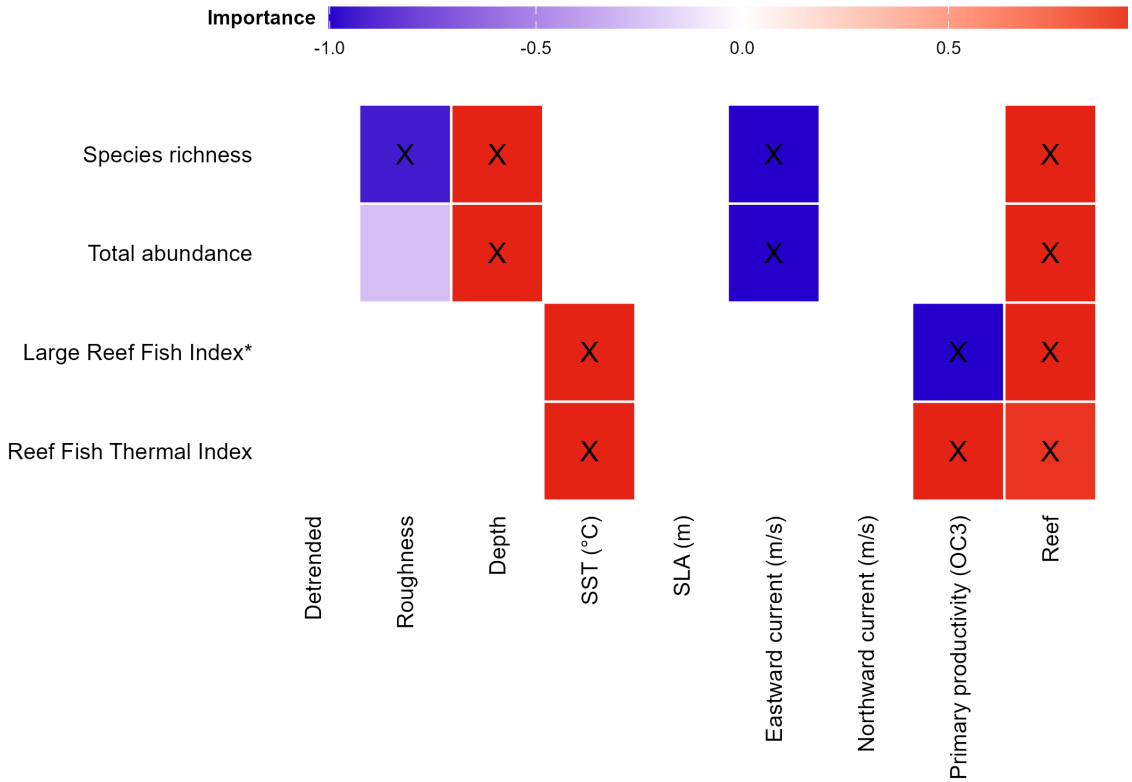


Figure C1.3: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

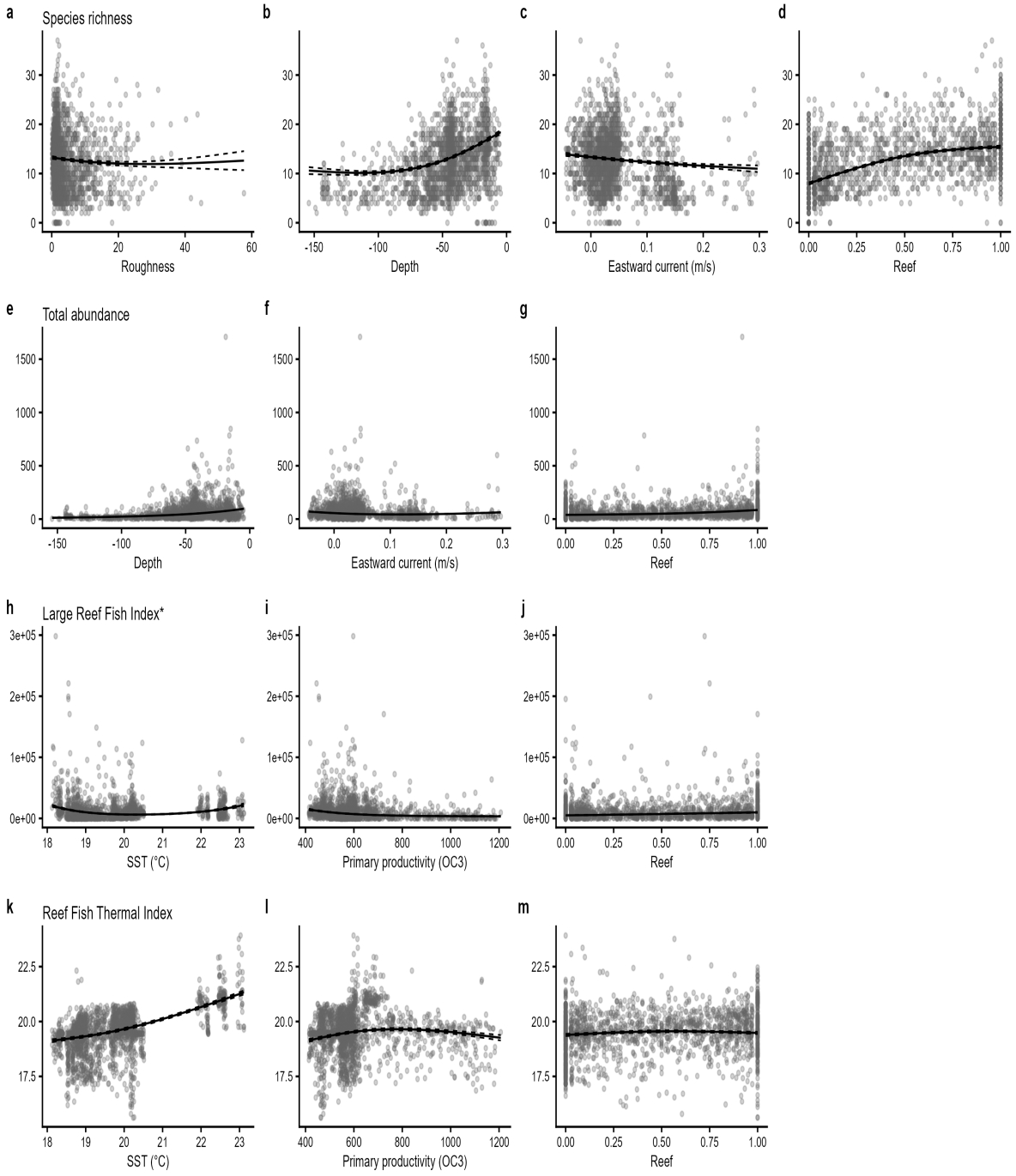


Figure C1.4: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of zone status

Table C1.3: Significance of the zone status term (No-Take against Fished) in each demersal fish GAM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary.

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Zone status: No-Take vs Fished	0.298	0.222	1.344	0.179
Total abundance	Zone status: No-Take vs Fished	0.217	0.006	35.340	<0.001
Large Reef Fish Index*	Zone status: No-Take vs Fished	0.206	0.057	3.595	<0.001
Reef Fish Thermal Index	Zone status: No-Take vs Fished	0.135	0.043	3.146	0.002